

- Min Zhang. 2025. Who wrote this? the key to zero-shot llm-generated text detection is geccore. In Proceedings of the 31st International Conference on Computational Linguistics, pages 10275–10292.
- Junchao Wu, Runzhe Zhan, Derek F. Wong, Shu Yang, Xinyi Yang, Yulin Yuan, and Lidia S. Chao. 2024. Detectrl: Benchmarking llm-generated text detection in real-world scenarios. In Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024.
- Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, Qingwei Lin, and Daxin Jiang. 2024a. Wizardlm: Empowering large pre-trained language models to follow complex instructions. In The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024. OpenReview.net.
- Haoyun Xu, Runzhe Zhan, Derek F. Wong, and Lidia S. Chao. 2024b. Let’s focus on neuron: Neuron-level supervised fine-tuning for large language model. CoRR, abs/2403.11621.
- Xianjun Yang, Wei Cheng, Yue Wu, Linda Ruth Petzold, William Yang Wang, and Haifeng Chen. 2024. DNA-GPT: divergent n-gram analysis for training-free detection of gpt-generated text. In The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024.
- Weichen Yu, Tianyu Pang, Qian Liu, Chao Du, Bingyi Kang, Yan Huang, Min Lin, and Shuicheng Yan. 2023. Bag of tricks for training data extraction from language models. In International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii, USA, volume 202 of Proceedings of Machine Learning Research, pages 40306–40320. PMLR.
- Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. Defending against neural fake news. In Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada, pages 9051–9062.
- Xiang Zhang, Junbo Jake Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In Advances in Neural Information Processing Systems 28: Annual Conference on Neural Information Processing Systems 2015, December 7-12, 2015, Montreal, Quebec, Canada, pages 649–657.
- Zheyuan Zhang, Fengyuan Hu, Jayjun Lee, Freda Shi, Parisa Kordjamshidi, Joyce Chai, and Ziqiao Ma. 2024. Do vision-language models represent space and how? evaluating spatial frame of reference under ambiguities. In The Thirteenth International Conference on Learning Representations.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. 2023. Representation engineering: A top-down approach to AI transparency. CoRR, abs/2310.01405.

A Appendix

A.1 Analysis of Activation Token

To investigate whether activation tokens contain specific tokens or which parts of speech enable the model to distinguish between HWT and LGT, We conducted an analysis of the word frequency and part-of-speech (POS) tags of Activation Tokens (the last 10% of tokens) and their relationship with RepreScore. The results are presented in Figure 8 and Figure 9. In general, the RepreScores for LGT are generally higher than those for HWT, with significant differences observed particularly in adjectives (ADJ), adverbs (ADV) and pronouns (PRON), while the differences for symbols (SYM) are the smallest. From the frequency distribution of the top 50 tokens, it is evident that the same token does not have identical RepreScore values in HWT and LGT; the scores for LGT are generally higher. This suggests that the RepreScore is not directly determined by the token itself. To explain this phenomenon, it is necessary to analyze from the perspective of Equation 1. Since the activation value of each token is computed based on the inputs of its preceding tokens, this implies that when the model processes a token t_n , it has already accounted for the contextual information from $\mathcal{T} = \{t_1, t_2, \dots, t_n\}$. Therefore, when we calculate the hidden representation changes of a token, these changes are essentially based on an analysis of the complete context rather than an isolated computation of the token itself.

A.2 Analysis of Model Memorization

Train ↓	Precision
ChatGPT	96.50
Llama-2-70b	96.50
Google-PaLM	95.55
Claude-instant	96.45
Mutil-LLMs	94.05
AVG.	95.81

Table 6: Precision of Different LLMs in Identifying HWT on 2,000 Samples from the Newly Collected Reddit Dataset Released in 2025. Precision is used as the dataset contains only a single class (HWT).

Previous research (Carlini et al., 2021) has demonstrated the potential to extract substantial portions of text from the training data of LLMs by employing carefully designed prompting techniques. This finding has been further substantiated

by subsequent work (Yu et al., 2023), which introduced advanced strategies for extracting training data. As a result, when text generated by LLMs is sourced directly from their training data, it becomes virtually indistinguishable from human-written text, rendering efforts to differentiate between LGT and HWT content effectively futile. Additionally, recent studies (Sun et al., 2025) have revealed that a significant portion of contemporary textual data now contains LGT while Reddit has exhibited relatively slower growth in this trend. To ensure that newly collected data consists exclusively of HWT, we utilize the latest Reddit dataset⁴ released in 2025, which contains content written after the training cut-off dates of the Llama-3.1-8B.⁵ The results in Table 6 demonstrate that the RepreGuard achieved exceptionally high precision across all model datasets when evaluating new HWT, with an average precision of 95.81%. This suggests that the RepreGuard is not influenced by model memorization, as the models do not simply recall the texts but can accurately identify the distinguishing features of LGT and HWT texts.

A.3 Performance After LGT Pretraining

As LLMs continue to evolve, an increasing proportion of their training data (Mukherjee et al., 2023; Xu et al., 2024a) is likely to consist of LGT, as the internet becomes increasingly saturated with content produced by these systems. This raises critical questions about the sustained effectiveness of RepreGuard when applied to large-scale datasets in such a scenario. To explore this, we curate a dataset comprising 1 million LGT to pre-train our surrogate model and systematically record the corresponding checkpoints, shown on the figure Figure 10. The results shown that As the proportion of LGT in pre-training increases from 0% to 100%, the AUROC of the RepreGuard exhibits a slight decline, yet overall performance remains robust. Specifically, the average AUROC decreases from 94.76% to 94.68%, demonstrating good overall robustness. In contrast, TPR@0.01 experiences a notable reduction, with the average TPR@0.01 decreasing from 80.92% to 73.72%. This suggests that pre-training with LGT diminishes the model’s detection capability under ex-

⁴https://huggingface.co/datasets/tensorshield/reddit_dataset_157

⁵<https://huggingface.co/meta-llama/Llama-3.1-8B>

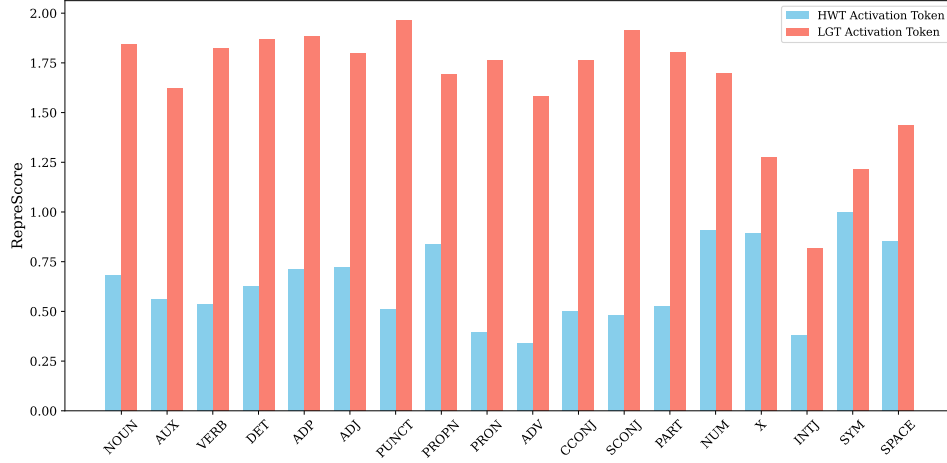


Figure 8: Average RepreScore values across different parts of speech (POS) tags for Activation Tokens.

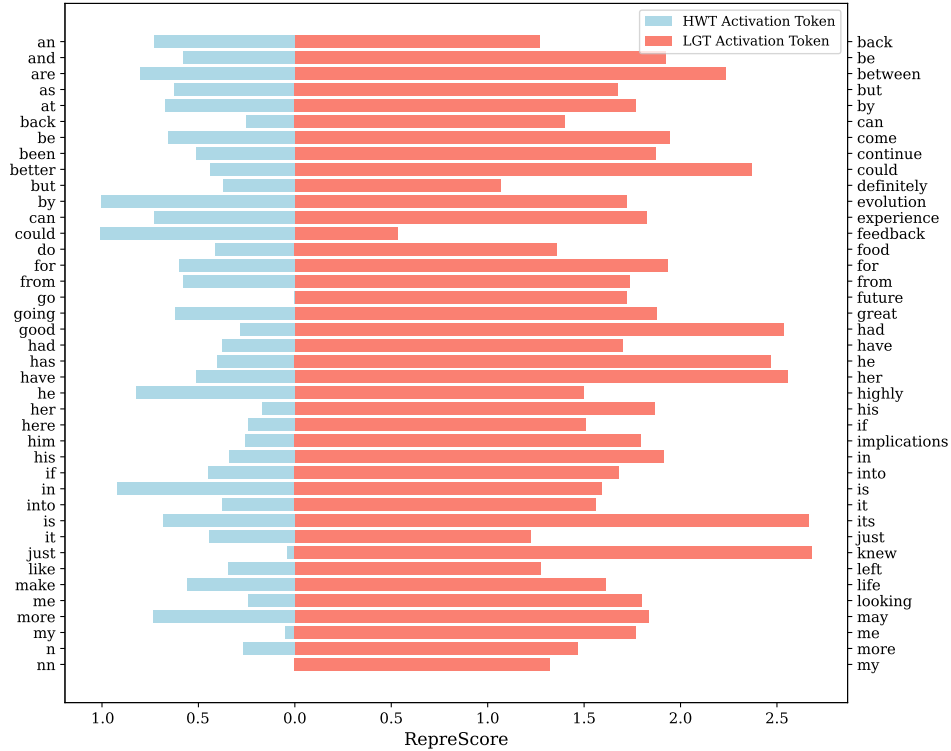


Figure 9: Frequency distribution of the top 50 Activation Tokens and their corresponding RepreScore values for HWT and LGT.

tremely low false positive rates.

A.4 Discussion on Hallucination Detection and LGT Detection using the Hidden Representation

Recent hallucination detection research has gradually shifted from focusing on the external performance to the internal hidden representation from LLMs. For instance, Masked Grouped Causal Tracing (MGCT) (Monea et al., 2024) reveals the

internal mechanisms underlying grounded and ungrounded behaviors by selectively perturbing and restoring hidden activations. Azaria and Mitchell (2023) used the hidden representation as feature inputs to train an external feedforward neural network classifier, enabling the automatic determination of statement veracity. The LLM-Check (Sriraman et al., 2024) method further extracts hidden representations during LLM response generation and calculates the covariance matrix (Hidden